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An experimental and kinetic modeling study of premixed nitromethane flames at low pressure

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Abstract

An experimental and modeling study is reported on three premixed nitromethane/oxygen/argon flames at low pressure (4.655 kPa) with equivalence ratios (ϕ) of 1.0, 1.5 and 2.0. Flame species were identified with tunable synchrotron vacuum ultraviolet photoionization. The mole fraction profiles of more than 30 species including intermediates and radicals were obtained by near-threshold burner scan with selected photon energies. With a Pt–6%Rh/Pt–30%Rh thermocouple, the flame temperature profiles were measured. A detailed kinetic mechanism with 88 species and 701 reactions was proposed and validated against experimental results of all three flames. The computed predictions showed satisfactory agreement with the experimental results. Based on a rate-of-production analysis, reaction pathway diagrams were obtained to describe the hydrocarbon oxidation process and nitrogenous species chemistry in the nitromethane flame with $\phi = 1.5$.
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Keywords: Nitromethane; Premixed flame; Kinetic modeling

1. Introduction

Nitromethane (CH_3NO_2) is known as a liquid explosive, an important solvent for chemical processing and analysis, and an additive to fuel for internal combustion (IC) engines as well [1]. It has a significant advantage over hydrazine due to reduced toxicity, and offers potential to serve as a candidate hydroxylammonium nitrate (HAN)-based monopropellant (HAN269MEO) [2]. Meanwhile, as the simplest CHON fuel containing nitro groups, nitromethane has been widely studied as a prototype for more complex energetic materials [3].

Previous studies on the process of nitromethane combustion focused mainly on its pyrolysis, both experimentally and numerically. In 1972, Glänzer and Troe carried out the first kinetic investigation of decomposition of highly diluted nitromethane via shock heating over the temperature range of 900–1400 K. They proposed a simplified mechanism involving only three reactions

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which described its pyrolysis [4]. In 1979, Perche et al. detected numerous secondary products of nitromethane pyrolysis including HCN and N_2 over the temperature range of 676–771 K in a static reaction vessel through periodically sampling and gas chromatography analysis [5–7]. While accounting for the time dependent concentrations, a mechanism involving 28 reactions was proposed. In 1985, Hsu and Lin performed an experimental investigation of nitromethane pyrolysis in a shock tube over the temperature range of 940–1520 K and the pressure range of 0.4–2 atm [8]. They employed a frequency stabilized continuous laser to monitor the appearance of NO and CO in the reflected shock region. A mechanism of 37 reactions was proposed as an extension of the work of Perche and coworkers. Zhang and Bauer studied the decomposition of nitromethane over the temperature range of 1000–1100 K in reflected shock waves with gas chromatography in 1997 [9]. They suggested that significant amounts of nitromethane were destroyed through secondary reactions that involved highly reactive radicals (H, OH and CH_3), and developed a mechanism of 99 reactions.

In 1998, Glarborg et al. developed a kinetic model of 438 reactions that describes C_1 and C_2 hydrocarbons/nitric oxide interactions under stoichiometric to fuel-rich conditions [10]. Shortly after that, Glarborg et al. proposed a comprehensive kinetic model of nitromethane thermal decomposition on the basis of previous results covering temperatures from 1000 to 1400 K and pressures of 0.5–6.0 atm [11]. As an extension of previous work, a nitromethane chemistry subset including 19 reactions were added. More recently, Boyer and Kuo developed a comprehensive model consisting of 47 species and 250 elementary reactions for predicting nitromethane flame structure and linear regression rate [3]. This mechanism was tested for its ability to capture the combustion behavior of nitromethane, but a detailed validation was not attempted.

More detailed experimental results about combustion intermediates in nitromethane flame did not appear until recently. Tian et al. performed an experiment on a premixed $CH_3NO_2/O_2/Ar$ flame at low pressure [12]. A mechanism developed from previous models of nitromethane decomposition [9] and nitromethane flame [3] involving 69 species and 314 reactions was proposed. However, Zhang and Bauer's model [9] is somewhat outdated, while Boyer and Kuo's model [3] is a simple high-pressure model. Therefore, both models are not very proper starting points for the modeling work of premixed nitromethane flames at low pressure. Besides, Tian's model was only validated against the measured concentrations of reactants and major products in a single nitromethane flame at $\phi = 1.5$, which is not adequate.

The present study is aimed at extending the experimental results of previous work on low-pressure premixed nitromethane flames, while developing a new detailed model to simulate the flame structure. Three premixed $CH_3NO_2/O_2/Ar$ flames at 4.655 kPa with equivalence ratios (ϕ) of 1.0, 1.5 and 2.0 were investigated with molecular-beam mass spectrometry and tunable synchrotron vacuum ultraviolet (VUV) photoionization. For each flame, mole fraction profiles of species identified have been determined, which help to validate the mechanism. The kinetic model in this work, involving 88 species and 701 reactions, was developed from a recent model that was validated against five low-pressure $NH_3/CH_4/O_2/Ar$ flames [13].

2. Experimental method

The experiments were performed at the National Synchrotron Radiation Laboratory (NSRL), Hefei, China. Detailed description of the instrument can be found elsewhere [14,15]. The temperature was measured with a Pt–6%Rh/Pt–30%Rh thermocouple of 0.1 mm in diameter, coated with Y_2O_3–BeO ceramic [16]. Radiation heat losses have been considered as well as the perturbation caused by the thermocouple in recalibration of the temperature profiles [17]. The uncertainty of temperature measurements were estimated to be ±100 K. The flame conditions are listed in Table 1. The equivalence ratio is calculated considering the oxidation reaction as $CH_3NO_2 + 1.25O_2 = CO_2 + NO + 1.5H_2O$. NO is assumed as the final NO_x product because it is more stable than NO_2 at high temperatures [10]. In each flame, Ar takes up 50% of the total volume flow rate. Here argon is used not only as the dilution gas but also as an inner standard for determination of flame species concentrations.

Flame species were identified by comparing the measured ionization thresholds to literature values. The errors of measured ionization energies are ±0.05 eV for species with high signal-to-noise (S/N) ratio and ±0.10 eV for those with low S/N ratio, considering the cooling effect of molecular beam. Photon energies of 16.77, 14.45, 13.62, 12.43, 11.94, 10.85, 10.05, 9.55 and 9.03 eV were selected in the burner scan to avoid fragmentation of intermediates and distinguish isomeric intermediates.

Table 1
Flame conditions.

ϕ	Total mass flow rate (g/cm^2 s)	Reactant mole fraction (%)		
		Ar	O_2	CH_3NO_2
1.0	0.002456	50.00	27.77	22.23
1.5	0.002541	50.00	22.73	27.27
2.0	0.002600	50.00	19.23	30.77

3. Kinetic modeling

The modeling study for the $CH_3NO_2/O_2/Ar$ flames was conducted using the Chemkin 2.0 Premix codes [18,19]. A starting mechanism was drawn from our recent study of NH_3-doped $CH_4/O_2/Ar$ flames [13]. This mechanism relies on recent work on oxidation of C_1/C_2-hydrocarbons [10,20], NH_3 [21] and HCN [22], as well as interactions of these components [10,23]. In the present work, the subsets for CH_3NO_2 and CH_3NO have been updated. Table 2 lists selected reactions of CH_3NO_2 subset. The full mechanism is available as the Supplementary data.

The nitromethane subset is based on the reaction mechanism for high-temperature conversion of CH_3NO_2 (>1000 K) proposed by Glarborg and co-workers [11,24]. The major consumption steps for nitromethane at high temperatures are thermal dissociation and reactions with the radical pool. The rate constant for the thermal dissociation (R1) was adopted from Glarborg et al. [11], who re-analyzed the experimental data from Glänzer and Troe [4] and other sources to provide an improved description of the pressure dependence of this step. Unimolecular rearrangement of nitromethane to methyl nitrite can be disregarded due to a high barrier [11]. Secondary reactions of CH_3NO_2 with the radical pool include the multi-product channel reaction $CH_3NO_2 + H$ to yield $CH_3 + HNO_2$ (R2), $CH_3NO + OH$ (R3) and $CH_2NO_2 + H_2$ (R4); as well as H abstraction from CH_3NO_2 by O (R5), OH (R6) and CH_3 (R9) to yield CH_2NO_2. These reactions have all been studied experimentally, but only low temperature data are available [11]. Note that Glarborg et al. [11] did not distinguish between the HONO and HNO_2 isomers. We assume that reaction (R2) forms HNO_2 rather than the more stable HONO species; this is consistent with the very low levels of HONO detected in the present flames. For the estimate of k_4, we maintain the analogy with $C_2H_6 + H$ but use an updated value. The $CH_3NO_2 + OH$ reaction is a multichannel reaction, but at low pressure the abstraction channel (R6) is believed to dominate [11]; the high-pressure channel to $CH_3OH + NO_2$ was excluded in the present work.

The CH_2NO_2 radical is believed to isomerize and dissociate into CH_2O and NO. The reaction occurs by ring closure to yield a three-membered ring isomer that will almost barrierlessly dissociate to products [25]. The current rate constant is based on the estimate for atmospheric pressure by Tricot et al. [26]. The activation energy is consistent with the recent estimate for the barrier by Liu et al. [25]. Even with the low A-factor assumed in the present study, this step remains the main consumption channel for CH_2NO_2. Reactions of CH_2NO_2 with the radical pool are included in the mechanism with estimated rates, mostly drawn from [11]. None of these steps are important for the predicted flame structure. For CH_3NO, the starting mechanism was extended with a subset drawn from the review of Dean and Bozzelli [27].

Table 2
Nitromethane reaction subset. Units are calories, cm^3, mol and s.

	Reaction mechanism	A	β	$E\alpha$	Source
1.	$CH_3NO_2(+M) = CH_3 + NO_2(+M)$	1.80E+16	0	58,500	[11]
	LOW/1.259E17 0 42000/				
	TROE/0.183 1E−30 1E30/				[11]
2.	$CH_3NO_2 + H = HNO_2 + CH_3$	3.30E+12	0	3730	[11], see text
3.	$CH_3NO_2 + H = CH_3NO + OH$	1.40E+12	0	3730	[11]
4.	$CH_3NO_2 + H = CH_2NO_2 + H_2$	4.90E+13	0	9220	est $1/2k_{C_2H_6+H}$
5.	$CH_3NO_2 + O = CH_2NO_2 + OH$	1.50E+13	0	5350	[11]
6.	$CH_3NO_2 + OH = CH_2NO_2 + H_2O$	5.00E+05	2	1000	[11]
7.	$CH_3NO_2 + HO_2 = CH_2NO_2 + H_2O_2$	3.00E+12	0	23,000	[11]
8.	$CH_3NO_2 + O_2 = CH_2NO_2 + HO_2$	2.00E+13	0	57,000	[11]
9.	$CH_3NO_2 + CH_3 = CH_2NO_2 + CH_4$	5.50E−01	4	8300	[11]
10.	$CH_3NO_2 + CH_3O = CH_2NO_2 + CH_3OH$	3.00E+11	0	7000	[11]
11.	$CH_3NO_2 + NO_2 = CH_2NO_2 + HONO$	3.00E+11	0	32,000	[11]
12.	$CH_2NO_2 = CH_2O + NO$	5.00E+11	0	36,000	[26], est
13.	$CH_2NO_2 + H = CH_3 + NO_2$	5.00E+13	0	0	[11]
14.	$CH_2NO_2 + O = CH_2O + NO_2$	5.00E+13	0	0	[11]
15.	$CH_2NO_2 + OH = CH_2OH + NO_2$	1.00E+13	0	0	[11]
16.	$CH_2NO_2 + OH = CH_2O + HONO$	1.00E+13	0	0	[11]
17.	$CH_2NO_2 + CH_3 = C_2H_5 + NO_2$	5.00E+13	0	0	est
18.	$CH_3ONO + H = CH_3OH + NO$	1.20E+11	0	1900	[40]
19.	$CH_3ONO + H = CH_2O + H_2 + NO$	1.40E+11	0	1900	[40]
20.	$CH_3ONO + O = CH_3O + NO_2$	1.40E+13	0	5210	[41]
21.	$CH_3ONO + OH = CH_3OH + NO_2$	6.00E+13	0	3505	[23]

4. Results and discussion

In this study, about 30 flame species were identified through comparison of the measured photoionization threshold to literature IEs from the NIST online database [28], including numerous intermediates which have relatively low concentrations but is potentially important for kinetic modeling. A detailed description of this method has been reported elsewhere [29].

Mole fraction profiles of identified species were calculated from experimental results of burner scan. A detailed description of the calculation method, which involves photoionization cross-section of flame species, has been reported previously [30]. While literature photoionization cross-sections of most species identified in this experiment are available [31–36], only few are available for nitrogenous intermediates. Thus an estimated value of 10 Mb was used when the photon energy was 0–0.95 eV higher than their respective IEs; and 20 Mb was used when the photon energy was 1–2 eV higher than the IEs [12]. The uncertainties of the mole fractions are 5–10% for major species, 25% for intermediates with known photoionization cross-sections, a factor of 2 for those with estimated photoionization cross-sections, and even larger for radicals due to their quenching effect on the probe surface [37].

Fig. 1 shows the temperature profiles of the three flames with shift downstream by 0.1, 0.2 and 0.2 cm, respectively, due to the cooling effect of the nozzle [38]. Generally, the peak position shifts downstream as ϕ increases, and the $\phi = 1.5$ flame has the highest temperature among the three flames.

The mole fraction profiles of the major species are shown in Fig. 2 with scattered symbols. In general, the profiles are shifted downstream with increasing ϕ. Fig. 2a and b present the mole fractions of reactants. In general, both reactants are consumed further downstream with ϕ increasing. However, in the post-flame region of the $\phi = 1$ flame, the O_2 concentration remains above zero (~4%) due to the impact of the relatively low temperatures upon oxidation reactions; CO and H_2 have also residual concentrations over this region. The mole fraction of argon is a qualitative reflection of the expansion in total amount of flame species. As is shown in Fig. 2c, for each flame the mole fraction profile of argon decreases within the same region of that of fuel consumption, and then tends to be flat. Compared with the other two flames, the decrease in Ar mole fraction in the $\phi = 1$ flame is smaller, which corresponds to the lower temperature at this equivalence ratio as expected. The increase in the CO peak mole fraction and the decrease in that of CO_2 displayed in Fig. 2d and e, respectively, can be attributed to the increasing C/O ratio. NO was detected as the major NO_x product, remaining with a high concentration and little changes in post-flame region of each flame, confirming its importance as a secondary product in nitromethane oxidation and its higher stability compared to NO_2 at relatively high temperatures. Fig. 2g shows that the mole fraction profiles of H_2O have quite similar tendency of that of temperature profiles, with the maximum value appearing in the $\phi = 1.5$ flame. The peak concentration of H_2, a common product in hydrocarbon flames, increases with increasing ϕ, shown in Fig. 2h.

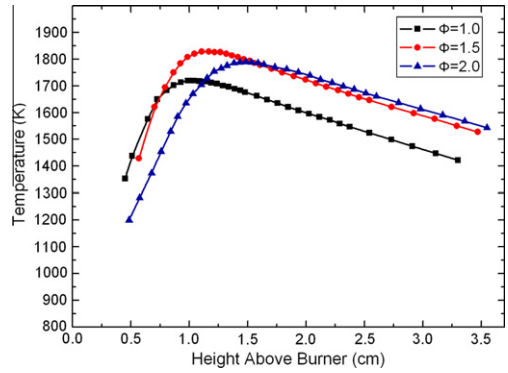

Fig. 1. Temperature profiles of all three nitromethane flames.

Modeling predictions of major species are presented in Fig. 2 with lines. The mechanism discussed in the section 'Kinetic modeling', and the temperature profiles presented in Fig. 1 were used in the modeling calculation. Due to the limitations in the temperature measurements, estimated temperatures at the burner surface of 800, 600 and 600 K were used for $\phi = 1.0$, 1.5 and 2.0 flames. In general, good agreement is observed between experimental and modeling results. However, the agreement for the $\phi = 1.0$ and 1.5 flames is better than that of the $\phi = 2.0$ flame. Uncertainties in the reaction mechanism are expected to contribute to the observed differences, particularly as the chemistry gets more complex with increasing ϕ. However, it should be noted also that the $\phi = 2.0$ flame is less stable than $\phi = 1.0$ or 1.5 flames, and thus more vulnerable to perturbations caused by the quartz nozzle, particularly when the sampling position is very close to the burner surface. Therefore the experimental uncertainties may rise in the $\phi = 2$ flame, possibly resulting in larger discrepancies between the experimental and modeling results. As for NO, the observed deviation may be within experimental uncertainty, considering the errors in measurements induced by the absence of a precise photoionization cross-section value of NO.

The experimental and predicted mole fractions of some selected intermediates are shown in Fig. 3. Similar to those of the major species, the mole fraction profiles are in general shifted

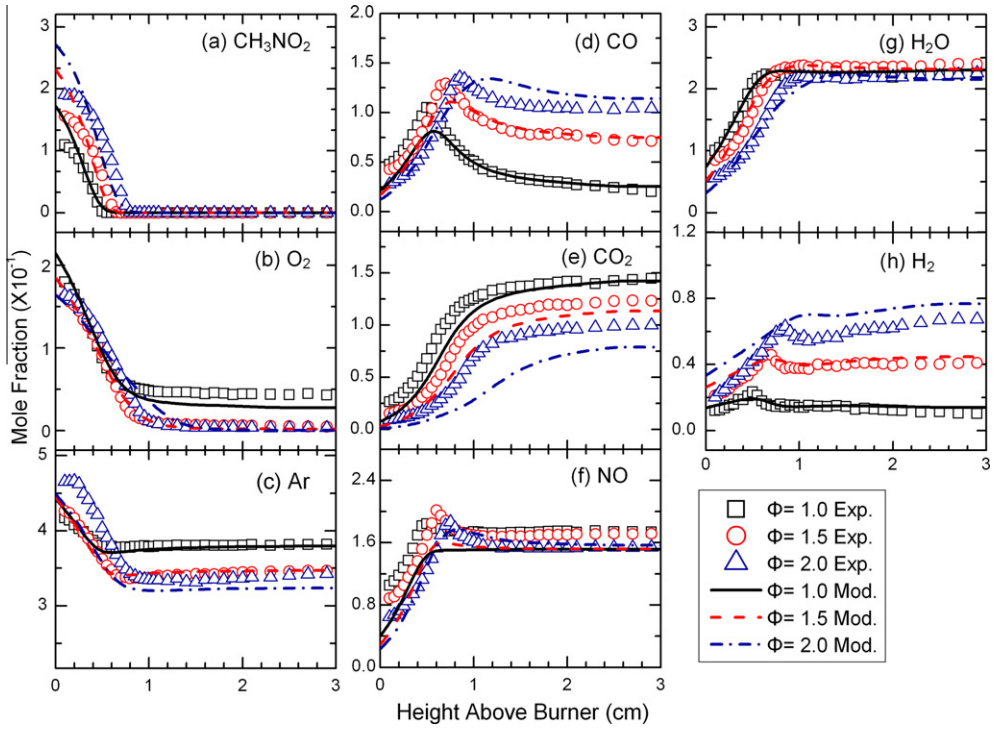

Fig. 2. Experimental and modeling mole fraction profiles of (a) CH_3NO_2, (b) O_2, (c) Ar, (d) CO, (e) CO_2, (f) NO, (g) H_2O and (h) H_2 in all three nitromethane flames, with scattered symbols for experimental results and lines for modeling.

downstream with increasing ϕ. The overall agreement achieved between experimental results and modeling predictions is satisfactory. To explain the reaction pathways in this work, we take the rate-of-production in $\phi = 1.5$ flame which is an integrated result along the flame as example. Reaction path diagrams for hydrocarbon oxidation and nitrogenous species conversion, respectively, for the $\phi = 1.5$ nitromethane flame are depicted in Fig. 4. Similar to previous studies on nitromethane pyrolysis [4], the dominant consumption step in the present flames is the C–N bond rupture producing CH_3 and NO_2, the mole fractions of which are shown in Fig. 3a and b. According to the result of the rate-of-production analysis, this reaction accounts for over 85% of the total consumption rate for nitromethane, with a minor contribution from secondary reactions of nitromethane with the radical pool, including H abstraction by H, OH and CH_3 to yield CH_2NO_2. The $CH_3NO_2 + H$ reaction, which has multiple product channels, also produces HNO_2 and CH_3NO; however, it accounts for less than 5% of the total consumption of nitromethane. The mole fractions of CH_3NO species shown in Fig. 3c is of the order of 10^{-4}, indicating its minor importance under the conditions of these experiments. The CH_2NO_2 radical mainly dissociates into $CH_2O + NO$, while CH_3NO yields CH_2NO prior to its oxidation by O_2 to produce CH_2O and NO_2. Thus the pyrolysis of nitromethane roughly produces two groups of species: hydrocarbons and nitric oxides, which lead to two reaction systems of hydrocarbon oxidation and nitrogenous species chemistry, respectively. The two systems are linked by numerous reactions of interaction between hydrocarbons and nitrogenous species. Species that play an important role in this process include hydrocarbon radicals such as CH_3, CH_2, CH and nitrogenous intermediates such as NO_2, NO and HNO.

The presence of even small amounts of nitrogen oxides can have a significant impact on fuel oxidation characteristics [23]. Since NO_2 and CH_3 are both directly produced from fuel pyrolysis in large amount, the reaction of $CH_3 + NO_2$ is greatly enhanced, which results in the promoted oxidation of CH_3 and the reduction of NO_2 to NO [23]. As can be seen from Fig. 3a and b, CH_3 has the highest mole fraction in $\phi = 1.5$ flame, while the mole fraction of NO_2 in $\phi = 1.5$ is obviously inhibited almost to the level of that in $\phi = 1.0$ flame. Rate-of-production analysis reveals that the reaction of CH_3 with NO_2 accounts for more than 40% of CH_3 consumption, producing CH_3O and NO. The oxidation process then goes through the major oxidation sequence of $CH_3O \rightarrow CH_2O \rightarrow HCO \rightarrow CO/CO_2$, with some branching reactions such as the production of CH_3OH and CH_2OH from CH_3O or directly from CH_3.

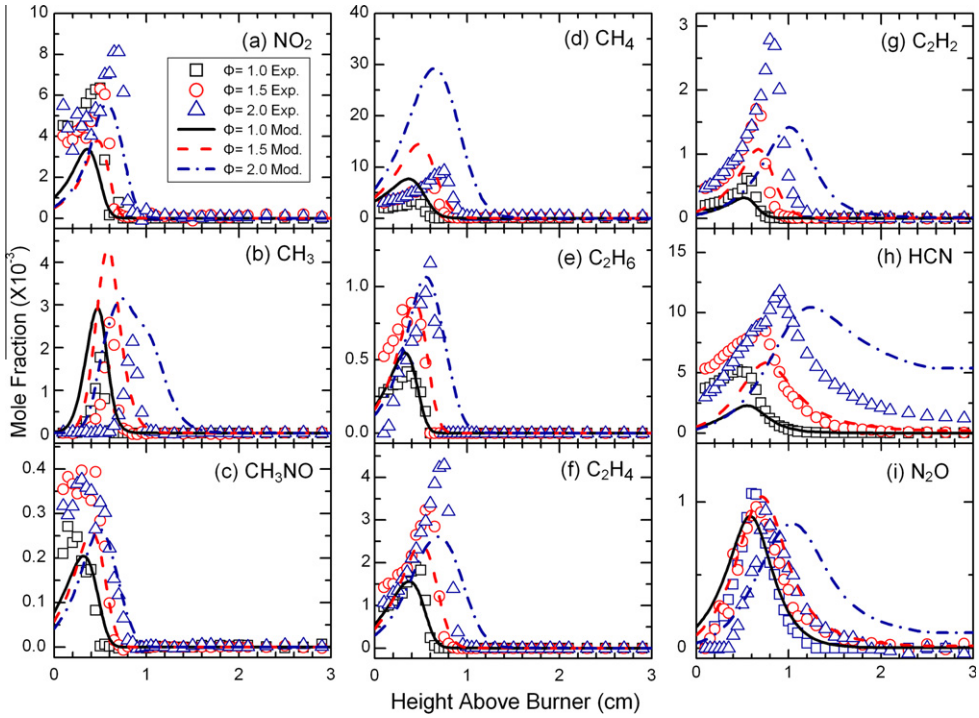

Fig. 3. Experimental and modeling mole fraction profiles of (a) NO_2, (b) CH_3, (c) CH_3NO, (d) CH_4, (e) C_2H_6, (f) C_2H_4, (g) C_2H_2, (h) HCN and (i) N_2O in all three nitromethane flames, with scattered symbols for experimental results and lines for modeling.

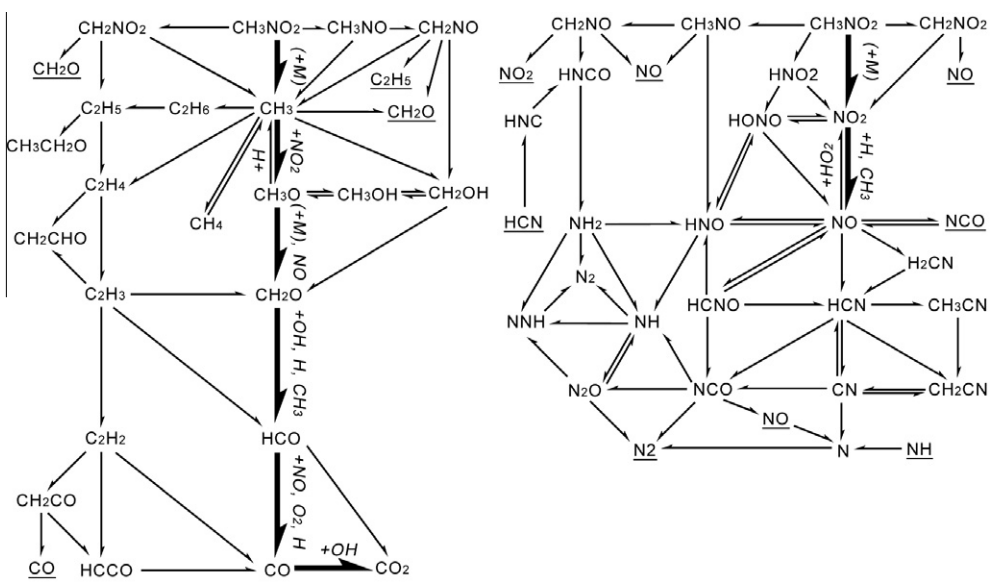

Fig. 4. Reaction pathway diagrams of hydrocarbon oxidation (on the left) and nitrogenous species chemistry (on the right) based on rate-of-production analysis of $\phi = 1.5$ nitromethane flame. Underlined species are duplicated ones for the neatness of diagram.

The mole fraction profile of CH_4 is shown in Fig. 3d. CH_4 is mainly produced in the reaction of $CH_3 + HNO$, which is a less important consumption pathway of CH_3 and another representative reaction of interaction between hydrocarbons and nitrogenous species, contributing up to ~60% of total CH_4 production. CH_4 is mainly recycled to CH_3 through the reaction with OH, H and O. In the $\phi = 2.0$ flame, the mole fraction of CH_4 is somewhat over-predicted, which requires further investigation. Feeding into C_2 hydrocarbon pool by self-recombination initiates another pathway for CH_3 consumption, leading to the oxidation sequence of $C_2H_6 \rightarrow C_2H_5 \rightarrow C_2H_4 \rightarrow C_2H_2 \rightarrow HCCO/CO$ with a few branching reactions and interactions with the major oxidation sequence. The peak values of the mole fraction profiles of C_2H_6, C_2H_4 and C_2H_2 have the same tendency to monotonously increase with ϕ, as shown in Fig. 3e–g, noting their peaks appear further downstream in the same order of oxidation. HCCO is mainly produced in the oxidation of C_2H_2 by O atoms, and consumed in the reaction with NO, yielding HCNO and CO.

For NO_2, 59% of total consumption is from the reaction with H atom. However, the reaction with CH_3 accounts for a comparable part of over 36%. While produced mostly from thermal dissociation of nitromethane, another part of NO_2 comes from the oxidation of NO by HO_2. It is noteworthy that this reaction also converts a relatively unreactive radical, HO_2, to a reactive one, OH, which promotes the hydrocarbon oxidation via formation of important chain-carrying radical [23].

NO plays an important role in many reactions with hydrocarbons, and all of its most important consumption pathways involve hydrocarbons. The reactions of NO with HCO and CH_3O producing HNO as the nitrogenous product account for over 70% of the total consumption. Only a small part (~5%) of the NO reacts with CH_2 to produce HCNO. However, HNO and HCNO are mainly recycled to NO in the reactions with the radical pool [10]. On the other hand, it is noteworthy that a minor part (2%) of NO is converted to HCN via the reactions with CH_i ($i = 1, 2, 3$) [39]. The mole fraction profiles of HCN are presented in Fig. 3h, which reach up to 10^{-2} magnitude in fuel-rich condition and remain a nonzero concentration over the post-flame region. The production of HCN from NO leads to a complex reaction system involving reactions mainly between HCN, NCO, HNCO, NH and NH_2. The interactions of these species with NO lead to formation of N_2, either directly or through N_2O or NNH. Fig. 3i shows the mole fraction profiles of N_2O, which plays an important role in this process, since the reaction of N_2O with H atom accounts for over 40% of N_2 production. Overall, it is seen that the chemistry of the present flames have implication for understanding and modeling the reburning process, in which nitric oxide is reduced by reaction with hydrocarbon radicals [10].

5. Conclusion

Three premixed nitromethane flames with $\phi = 1.0$, 1.5 and 2.0 at low pressure have been investigated experimentally with molecular-beam mass spectrometry and synchrotron VUV photoionization techniques. More than 30 flame species have been identified with their mole fraction profiles calculated as a function of HAB.

A kinetic model involving 88 species and 701 reactions has been developed and validated against the experimental results of all three flames. The prediction of the mechanism is generally in reasonable agreement with the experimental observations for the radicals and intermediates which are of significance in nitromethane combustion. Based on the analysis of rate-of-production, the reaction systems for hydrocarbon oxidation and nitrogenous species chemistry are obtained with links based on numerous reactions that involve species such as hydrocarbon radicals and nitric oxides. Special emphasis has been put on the key reactions between hydrocarbons and nitrogenous species.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.proci.2010.06.002.

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